

Drawings and PCB Photos - Automated Part Costing

Problem Statement

PII:

No

- Most of what a cost engineer needs is written on the 2D drawing, not held in the 3D model. Tolerances, surface finish, heat treatment and material notes are read by eye and typed in by hand.
- Many parts never reach us as a 3D model. We are sent a drawing. For electronics we are sent a photograph of the board.
- The costing method and the rate data do not change. This is only about reading the input.

Projected ROI

Cost: £???

Return: £???

What is the return based on?

Parts we cannot cost today, plus the hours spent reading drawings by hand.

Time (hrs/week)

Current State

- A cost engineer reads the drawing, notes the tolerances, finish and heat treatment, and types them into CAPPe alongside the figures taken off the model.
- For a circuit board, someone lists every component by hand from a photograph or a sample board, then prices each line before any costing can begin.
- Because it takes so long, we cost the parts we are asked about rather than the ones worth asking about, and a cost is rarely revisited when the design moves on.

Proposed Solution/Model

- Conduct a 3-stage Proof of Value (PoV) project, run the same way as the CAD to Cost PoV.
- Use an assistant to read the 2D drawing and the board photo and write down what it found — tolerances, finish and heat treatment for a part, the component list for a board.
- Everything it reads is checked against the measured geometry or a price catalogue before it is used. It reads the input. It does not set any price.
- Generate cost estimates for a sample of parts and boards, and validate outputs vs existing cost estimates.

Expected Benefits:

- **Wider Coverage:** Cost the parts and boards we cannot cost today, because a drawing or a photograph is all we have.
- **Improved Quality:** Fewer mistakes copying figures across, and the same drawing read the same way every time.

Technical Scoping:

Enabling AI on an existing system?
(YES)

The reading is done inside CostVision. CAPPe is not changed.

Model used: Claude (Anthropic)

Sonnet and Haiku to read, Opus where a board needs a closer look.

Technical implementation/support required?
(YES)

It needs an approved route to the model. Option 2 runs with that route switched off. This one cannot.